

Lessons

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Multiplying and Dividing Number... ▾

Multiplying and Dividing Numbers with Signs

Multiplication is fast addition. In other words, you can multiply when you have to add the same number many times.

$5 + 5$ is the same as 2×5 . Both equal 10.

$(-5) + (-5)$ is the same as $2 \times (-5)$. Both equal -10 .

Multiplying a negative number by a negative number is hard to visualize. You can apply your reasoning skills to find out how it works.

What number works in the blank? $-5 \times \underline{\hspace{1cm}} = 10$

If you put a positive number in the blank, the product would have to be -10 . A positive number won't work. The missing number must be negative: $-5 \times -\mathbf{2} = 10$

Remember that multiplication can be written in three ways: using a times symbol ($\times$), using a raised dot ($\bullet$), or writing the numbers next to each other in parentheses.

In algebra, division is written using a fraction bar. For example, $15 \div 5$ is written $\frac{15}{5}$.

📌 Tip

Use these rules to multiply or divide two signed numbers:

If the signs are the same, the answer is positive. $3 \times 4 = 12$ $-3 \times -4 = 12$

$\frac{3}{4} = 0.75$

$\frac{-3}{-4} = 0.75$

If the signs are different, the answer is negative. $-3 \times 4 = -12$ $3 \times -4 = -12$

$\frac{3}{-4} = -0.75$

$\frac{-3}{4} = -0.75$

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Lesson 2: Operations with Whole Numbers and Decimals

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